SULFUR 13

Sulfur (sometimes spelt *sulphur*) is a yellow, brittle solid at room temperature. It doesn't dissolve in water but is soluble in some organic (carbon-based) solvents such as methylbenzene. Being a typical non-metal, sulfur does not conduct electricity and has a low melting point (approx 115°C). Sulfur was called **brimstone** in old books. It occurs as the almost pure element in underground deposits (e.g. those found in Sicily and the USA). It is the sixteenth most-abundant element in the Earth's crust.

Sulfur is in Group VI in the Periodic Table and so sulfur atoms have six electrons in the outer (valence) shell. Figure 1 shows how the electrons are arranged in an atom of sulfur. Sulfur forms many compounds; e.g. oxides, sulfides, sulfites and sulfates. Sulfur dioxide (SO_2), iron (II) sulfide (FeS), sodium sulfite (Na_2SO_3) and copper sulfate ($CuSO_4$) are common examples of sulfur compounds.

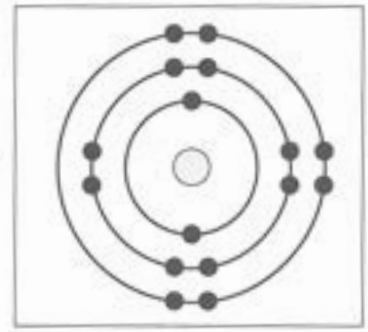


Figure 1

The most important use of sulfur is in the manufacture of sulfur compounds such as sulfuric acid (H_2SO_4) which is a very important industrial chemical. It is important in medicine for '*sulfa*' (sulphonamide) drugs and in many skin treatments. Sulfur is also a raw material in the manufacture of matches, rubber tyres, dyes and gunpowder.

When sulfur burns in air, it gives a blue flame, forming sulfur dioxide. This is a colourless, heavier-than-air gas. Sulfur dioxide is also formed when fossil fuels such as coal and oil are burned. Coal usually contains 0.5%–5% sulfur. Sulfur dioxide has a very irritating smell and is poisonous even at low levels. It is thought to contribute to bronchitis and other lung diseases. Sulfur dioxide is acidic and has bleaching properties. This compound can be used to kill germs and, at low concentrations, as a food preservative, as it prevents the growth of moulds and bacteria.

Millions of tonnes of sulfur dioxide are emitted every year into the atmosphere by power stations. The sulfur dioxide reacts with water vapour, oxygen and forms sulfuric acid which then falls as **acid rain** – damaging buildings and killing trees and fish in lakes. Sulfur dioxide is also produced when some metals are extracted from certain ores; e.g. zinc from zinc blende (ZnS), iron from iron pyrites (FeS_2) and lead from galena (PbS). Large amounts of **quicklime** (calcium oxide, CaO) have been added to acidic lakes to neutralise the acidity.

World production of sulfuric acid is in the region of hundreds of millions of tonnes annually. The largest single use of sulfuric acid is in the manufacture of fertilisers. It is also used in making organic products, refining petroleum (crude oil), treating metals and making paints, pigments and artificial fabrics such as rayon. One of the more commonly known uses of sulfuric acid is in the car battery or **lead-acid accumulator**.

1 Give three characteristics of sulfur that indicate it is a non-metal.

2 Name another element that is in Group VI of the Periodic Table.

3 How many electrons does an atom of sulfur have in total?

4 In chemistry terms, what does *organic* mean?

5 Write the name and formula of each sulfur compound mentioned on this sheet.

6 What do you think the formula of sulfur trioxide is?

7 Give three products that use sulfur as a raw material in their production.

8 A label on a wine bottle said that a very small quantity of sulfur dioxide had been added. Why was this done?

9 Name one industry that is a major source of the sulfur dioxide in the atmosphere.

10 Name two chemicals present in a car battery.

11 Give two uses of sulfuric acid.

Research 12 Sulfur can exist as allotropes. What is an *allotrope*?

© Peter Gribben, Wendy St Germain and Blake Education 2005

© Blake Education ISBN 978 1 92561 067 3 Assignments in Chemistry Book 3

The Frasch process

Sulfur can be extracted as a by-product from refining petroleum and the processing of metal ores that contain sulfur. Most sulfur is obtained by an unconventional mining method called the Frasch process. This technique was invented in 1891 by Herman Frasch and relies on the low melting point of sulfur and its insolubility in water.

Figure 2 shows a simplified version of the mining operation.

- There are three pipes, stacked inside one another.
- These pipes are sunk into a deposit of sulfur.
- The outer pipe carries superheated steam (160°C) to the sulfur deposit.
- The sulfur is melted by the steam.
- The central pipe carries compressed air to the mixture of molten sulfur and steam.
- The compressed air forces the sulfur/water mixture to the surface via the intermediate pipe.
- On the surface, the mixture is collected and cools down. The sulfur solidifies and separates out.

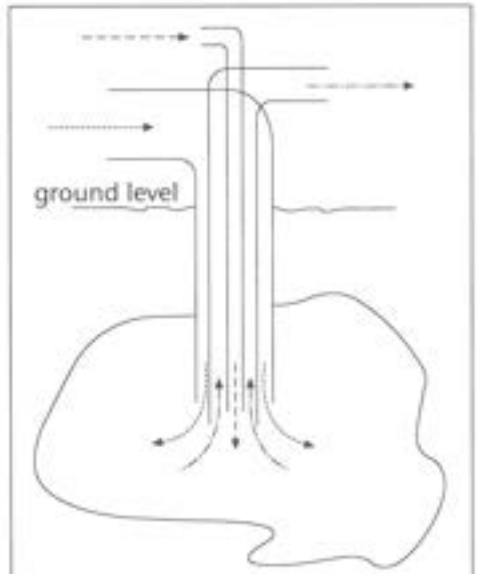


Figure 2

13 Copy figure 2. Use the information above to label your diagram.

14 Which physical properties of sulfur allow the Frasch process to work?

The Contact process

Sulfuric acid is one of the world's most important chemicals and over 50 million tonnes is produced annually. About two-thirds of the sulfuric acid produced is used in the production of fertilisers. It is also used in the production of dyes and explosives and in the petrochemical industry. The Contact process is the major industrial method of producing sulfuric acid. The first stage in the Contact process is to produce sulfur dioxide (SO_2). This can be done by burning sulfur in air or by roasting metal sulfides.

15 Write a balanced equation for the formation of sulfur dioxide by the combustion of sulfur.

16 Write a balanced equation for the formation of sulfur dioxide by the reaction between oxygen and iron (II) sulfide (FeS) also forming iron (III) oxide (Fe_2O_3).

The next step is to oxidise sulfur dioxide to sulfur trioxide (SO_3) using vanadium (V) oxide as the catalyst at about 450°C.

17 Write a balanced equation for the formation of sulfur trioxide by the reaction between sulfur dioxide and oxygen.

Sulfur trioxide gas is passed through concentrated sulfuric acid (H_2SO_4) and an unstable substance called oleum ($H_2S_2O_7$) is formed (*oleo*, meaning oil). When water is added to oleum, sulfuric acid is formed.

18 Write a balanced equation for the formation of oleum by the reaction between sulfur trioxide and concentrated sulfuric acid.

19 Write a balanced equation for the reaction between water and oleum.

20 The chemists in charge of plants where the Contact process is carried out must ensure that sulfur dioxide does not escape into the atmosphere. Give a reason for this.

21 Name the major use of sulfuric acid.

22 Draw up a flow chart to show the Contact process.

23 Mark each statement as true or false.

a Hydrochloric acid is produced by the Contact process.

b Sulfur is soluble in water.

c Most sulfuric acid is used in the production of fertilisers.

d Sulfur is a by-product when petroleum is refined.

© Peter Gribben, Wendy St Germain and Blake Education 2005
© Blake Education ISBN 978 1 92561 067 3 Assignments in Chemistry Book 3